

Minimum Substring

Hectron

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1 Description

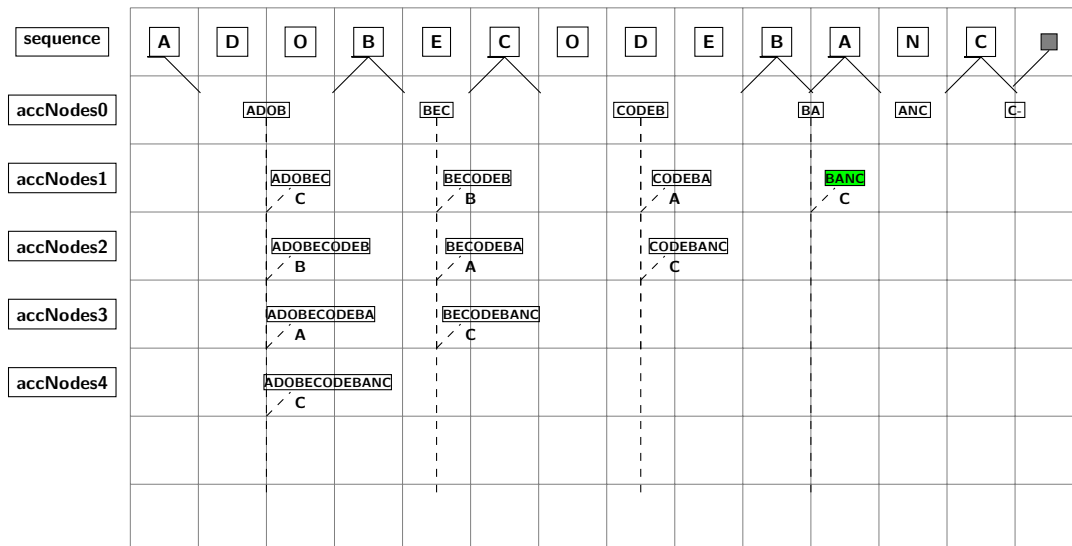
Given two strings s and t of lengths m and n respectively, return the minimum window of s such that every character in t (including duplicates) is included in the window.

- **Input:**
 $s = \text{"ADOBECODEBANC"}$
 $t = \text{"ABC"}$
- **Output:**
 "BANC"
- **Explanation:**
The minimum window substring "BANC" includes 'A', 'B', and 'C' from string t .

2 Function/Method Declaration

```
class Solution {
public:
    std::string minWindow(std::string s, std::string t);
}
```

3 How to solve



Search for minimum string requires runtime of $O(n \log_2(n))$.

Figure 1: Search for minimum substring example.